MUHAMMAD ZAIN UL ABEDIN

MECHANICAL ENGINEER | VISUAL DATA ANALYSIS & ASSET INTEGRITY

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PROFILE

Mechanical Engineering graduate focused on visual data analysis, condition assessment and asset integrity management for onshore and offshore infrastructure. Developed AssetInspect, an independent portfolio project demonstrating manual data annotation, mechanical, structural and hydraulic feature identification, basic quality assurance, data analytics and engineering inspection reports using synthetic evidence.

SKILLS

Inspection and assessment: Visual data analysis; manual data annotation; fault and defect identification; condition assessment; asset integrity management; basic quality assurance; technical report writing.

Engineering drawings and 3D: P&IDs; isometrics; engineering schematics; process diagrams; CAD models of piping and structures; 3D model review; point cloud data; basic metrology using model coordinates.

Software familiarity: Autodesk AutoCAD, Autodesk Navisworks and Autodesk ReCap; MS Excel; Google Sheets; Google Workspace (G-Suite); MS Office; Linux.

Standards familiarity: API, NACE, ASCE and ASME inspection and coating standards; client-defined assessment criteria.

Data and collaboration: Data analytics; Python; Django; JavaScript; Git/GitHub; hydraulics; fluid mechanics; attention to detail; analytical problem solving; effective communication; teamwork; adaptability; continuous improvement.

PROJECT

AssetInspect - Visual Infrastructure Inspection

Independent portfolio project | Django application and standalone offline HTML demo

Technologies: Python, Django, JavaScript, HTML/CSS, SQLite, IndexedDB, Pillow and ReportLab

- **Visual inspection and annotation:** Organized synthetic pipeline, tank and marine-structure evidence in asset/inspection registers; added JPEG/PNG upload and mouse, touch and keyboard rectangle annotation.
- **Faults and engineering features:** Separated defect indications from mechanical, structural and hydraulic features, with component labels, equipment tags, preliminary severity, observations and recommended follow-up.
- **Drawings and assessment:** Supported raster P&IDs, isometrics, schematics and CAD screenshots; recorded client criteria, condition/action, rationale, limitations, follow-up dates and API/NACE/ASCE/ASME reference metadata.
- **Quality assurance and review:** Added inspector/reviewer QA checklists, overlap hints, review comments, change requests, resubmission, approval and audit history; inspection edits invalidate prior QA.
- **3D and point cloud data:** Imported OBJ, ASCII PLY and XYZ/CSV geometry for rotate/pan/zoom review and two-point distances in declared model units; retained endpoints, source and accuracy limitations.
- **Data analytics and spreadsheets:** Created searchable, filtered and paginated worklists for up to 1,000 annotations, asset/component/severity summaries, CSV for Excel/Google Sheets and a four-sheet Excel XML workbook.
- **Engineering reporting:** Generated inspection/condition assessment PDF reports with annotated evidence, observations, assessment rationale, QA, model-coordinate measurements and recommended follow-up.
- **Delivery and verification:** Implemented authenticated roles and permission checks in Django; offline HTML uses simulated roles, local persistence, JSON backups and shareable copies. Verified with 47 Django tests and offline browser checks.

EDUCATION

Bachelor's Degree in Mechanical Engineering | Completed 2019

International Islamic University Islamabad | CGPA: 3.57/4.00

Intermediate (Pre-Engineering) | Punjab Group of Colleges, Islamabad | 2015 | 82%

Matriculation (Science) | Islamabad Model College for Boys G-11/1, Islamabad | 2013 | 86%

CERTIFICATIONS

Pakistan Engineering Council

Generative AI Application Developer | The Role of Engineers in Safe and Sustainable Operations | Professional Ethics